Unit: Unit 0: Foundations of Algebra
Topic: Divisibility Division Algorithm

Name: ____________________
Date: ____________

Challenge (Habanero)

Q1. A gamer has 48 hours of gameplay data. If they want to divide it into segments of 8 hours, how many segments can they make?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 4

Q2. A coding error occurs every 15 lines of code. If a programmer wrote 135 lines, how many errors will occur?
(A) 8
(B) 9
(C) 10
(D) 11
(E) 7

Q3. A gamer scored 945 points. If they want to know how many sets of 15 points they have, what is the quotient?
(A) 62
(B) 63
(C) 60
(D) 61
(E) 59

Q4. A data packet is 1200 bits long. If it's divided into packets of 8 bits, what is the remainder?
(A) 0
(B) 4
(C) 2
(D) 1
(E) 3

Q5. If a gamer has 378 gems and wants to put them into boxes of 9 gems, how many boxes can they fill?
(A) 41
(B) 42
(C) 40
(D) 43
(E) 39

Q6. A bug occurs every 12 lines of code. If 84 lines of code were written, how many bugs will occur?
(A) 6
(B) 7
(C) 8
(D) 9
(E) 5

Q7. A gaming session lasted 288 minutes. If it was divided into segments of 12 minutes, how many segments were there?
(A) 23
(B) 24
(C) 22
(D) 25
(E) 21

Q8. A gamer has 630 coins and wants to put them into piles of 10 coins. How many piles can they make?
(A) 62
(B) 63
(C) 60
(D) 61
(E) 59

Open Ended Questions (FRQ)

Q9. A gaming company has 864 megabytes of data to process. If they process it in batches of 12 megabytes, how many batches can they process and what is the remainder? Show your work and express your answer as $\frac{a}{b}$, with the remainder in the numerator and the divisor in the denominator.

Q10. A programmer is debugging a code with 150 lines of code, and they want to check every 5 lines. How many times will they check the code, and what is the remainder? Express the answer in the form $a \cdot b + c$ where a is the quotient, b is the divisor, and c is the remainder.

Chef's Tips

- Check the divisibility rules
- Perform long division when necessary
- Express answers with remainders correctly